

Lecture 4: Properties of OLS

Economics 326 — Introduction to Econometrics II

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The OLS estimators are random variables

- The model

$$Y_i = \alpha + \beta X_i + U_i, \\ E(U_i | X_1, \dots, X_n) = 0.$$

Conditioning on X in $E(U_i | X_1, \dots, X_n) = 0$ allows us to treat all X 's as fixed, but Y is still random.

- The estimators

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \text{ and } \hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

are random because they are functions of random data.

The estimators are linear

- Since

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2},$$

we can write $\hat{\beta} = \sum_{i=1}^n w_i Y_i$, where

$$w_i = \frac{X_i - \bar{X}}{\sum_{l=1}^n (X_l - \bar{X})^2}.$$

After conditioning on X 's, w_i 's are not random.

- For $\hat{\alpha}$,

$$\begin{aligned} \hat{\alpha} &= \bar{Y} - \hat{\beta} \bar{X} \\ &= \frac{1}{n} \sum_{i=1}^n Y_i - \left(\sum_{i=1}^n w_i Y_i \right) \bar{X} \\ &= \sum_{i=1}^n \left(\frac{1}{n} - \bar{X} w_i \right) Y_i \\ &= \sum_{i=1}^n \left(\frac{1}{n} - \bar{X} \frac{X_i - \bar{X}}{\sum_{l=1}^n (X_l - \bar{X})^2} \right) Y_i. \end{aligned}$$

Unbiasedness

- $\hat{\beta}$ is called an unbiased estimator if $E\hat{\beta} = \beta$.

- Suppose that $Y_i = \alpha + \beta X_i + U_i$, $E(U_i | X_1, \dots, X_n) = 0$. Then $E\hat{\beta} = \beta$.

$$\begin{aligned}
\hat{\beta} &= \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= \frac{\sum_{i=1}^n (X_i - \bar{X}) (\alpha + \beta X_i + U_i)}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= \alpha \frac{\sum_{i=1}^n (X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2} + \beta \frac{\sum_{i=1}^n (X_i - \bar{X}) X_i}{\sum_{i=1}^n (X_i - \bar{X})^2} + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= \alpha \frac{0}{\sum_{i=1}^n (X_i - \bar{X})^2} + \beta \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2} + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}.
\end{aligned}$$

- or

$$\hat{\beta} = \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Unbiasedness

- Once we condition on $X_1, \dots, X_n$, all X 's in

$$\hat{\beta} = \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

can be treated as fixed.

- Thus,

$$\begin{aligned}
E(\hat{\beta} | X_1, \dots, X_n) &= E\left(\beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \mid X_1, \dots, X_n\right) \\
&= \beta + E\left(\frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \mid X_1, \dots, X_n\right) \\
&= \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) E(U_i | X_1, \dots, X_n)}{\sum_{i=1}^n (X_i - \bar{X})^2}.
\end{aligned}$$

Unbiasedness

- Thus, with $E(U_i | X_1, \dots, X_n) = 0$, we have

$$\begin{aligned}
E(\hat{\beta} | X_1, \dots, X_n) &= \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) E(U_i | X_1, \dots, X_n)}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) \cdot 0}{\sum_{i=1}^n (X_i - \bar{X})^2} = \beta.
\end{aligned}$$

- By the LIE, $E\hat{\beta} = E[E(\hat{\beta} | X_1, \dots, X_n)] = E[\beta] = \beta$.

Strong exogeneity of regressors

- The regressor X is strongly exogenous if $E(U_i | X_1, \dots, X_n) = 0$.
- Alternatively, we can assume that $E(U_i | X_i) = 0$ and all observations are independent:

$$\begin{aligned}
E(U_1 | X_1, \dots, X_n) &= E(U_1 | X_1), \\
E(U_2 | X_1, \dots, X_n) &= E(U_2 | X_2) \text{ and etc.}
\end{aligned}$$

- The OLS estimator is in general biased if the strong exogeneity assumption is violated.

Variance of $\hat{\beta}$

- If $Y_i = \alpha + \beta X_i + U_i$, $E(U_i | X_1, \dots, X_n) = 0$, and

$$E(U_i^2 | X_1, \dots, X_n) = \sigma^2 = \text{constant},$$

and for $i \neq j$

$$E(U_i U_j | X_1, \dots, X_n) = 0,$$

then

$$\text{Var}(\hat{\beta} | X_1, \dots, X_n) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- The assumption $E(U_i^2 | X_1, \dots, X_n) = \sigma^2 = \text{constant}$ is called (conditional) homoskedasticity.
- The assumption $E(U_i U_j | X_1, \dots, X_n) = 0$ for $i \neq j$ can be replaced by the assumption that the observations are independent.

Variance of $\hat{\beta}$

$$\text{Var}(\hat{\beta} | X_1, \dots, X_n) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- The variance of $\hat{\beta}$ is positively related to the variance of the errors $\sigma^2 = \text{Var}(U_i)$.
- The variance of $\hat{\beta}$ is smaller when X 's are more dispersed.

Variance of $\hat{\beta}$

- We are going to condition on X 's and will treat them as constants. All expectations below are implicitly conditional on X 's.
- We have $\hat{\beta} = \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}$ and $E\hat{\beta} = \beta$.

$$\begin{aligned} \text{Var}(\hat{\beta}) &= E\left[(\hat{\beta} - E\hat{\beta})^2\right] \\ &= E\left[\left(\frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}\right)^2\right] \\ &= \left(\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2}\right)^2 E\left[\left(\sum_{i=1}^n (X_i - \bar{X}) U_i\right)^2\right]. \end{aligned}$$

Variance of $\hat{\beta}$

- Expanding the square,

$$\begin{aligned} \left(\sum_{i=1}^n (X_i - \bar{X}) U_i\right)^2 &= \sum_{i=1}^n \sum_{j=1}^n (X_i - \bar{X})(X_j - \bar{X}) U_i U_j \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 U_i^2 + \sum_{i=1}^n \sum_{j \neq i}^n (X_i - \bar{X})(X_j - \bar{X}) U_i U_j. \end{aligned}$$

- Since $E(U_i U_j) = 0$ for $i \neq j$,

$$\begin{aligned} E\left[\left(\sum_{i=1}^n (X_i - \bar{X}) U_i\right)^2\right] &= \sum_{i=1}^n (X_i - \bar{X})^2 E U_i^2 + 0 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 \sigma^2. \end{aligned}$$

Variance of $\hat{\beta}$

We have

$$\begin{aligned} \text{Var}(\hat{\beta}) &= \left(\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2} \right)^2 E \left[\left(\sum_{i=1}^n (X_i - \bar{X}) U_i \right)^2 \right], \\ E \left[\left(\sum_{i=1}^n (X_i - \bar{X}) U_i \right)^2 \right] &= \sigma^2 \sum_{i=1}^n (X_i - \bar{X})^2, \end{aligned}$$

and therefore,

$$\begin{aligned} \text{Var}(\hat{\beta}) &= \left(\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2} \right)^2 \sigma^2 \sum_{i=1}^n (X_i - \bar{X})^2 \\ &= \left(\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2} \right) \sigma^2. \end{aligned}$$

Normality of $\hat{\beta}$

- Assume that U_i 's are jointly normally distributed conditional on X 's.
- Then $Y_i = \alpha + \beta X_i + U_i$ are also jointly normally distributed.
- Since $\hat{\beta} = \sum_{i=1}^n w_i Y_i$, where $w_i = \frac{X_i - \bar{X}}{\sum_{i=1}^n (X_i - \bar{X})^2}$ depend only on X 's, $\hat{\beta}$ is also normally distributed conditional on X 's.
- Conditional on $X_1, \dots, X_n$

$$\begin{aligned} \hat{\beta} &\sim N(E\hat{\beta}, \text{Var}(\hat{\beta})) \\ &\sim N\left(\beta, \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}\right). \end{aligned}$$